ECEN-662: Homework 4

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Instructor: Dr. Tie Liu

Brody Jordan
431005357

Problem 1

Let $\theta > 0$ be an unknown parameter, and let $X = (X_0, X_1, \ldots, X_n)$ where $X_0 \sim Poisson(\theta)$ and for any $i \in [n]$,

$$X_i | (X_0, X_1, \ldots, X_{i-1}) \sim Poisson(\theta X_{i-1}).$$

(i) Find the MLE of θ based on the observation X.

(ii) Compute $I(\theta)$ with respect to the observation X.

Solution

Part (i):

$$\begin{aligned}
\mathcal{L}(\theta) &= \log P(X_0) + \sum_{i=1}^{n} \log P(X_i | X_0, X_1, \ldots, X_{i-1}) \\
&= [-\theta + X_0 \log\theta - \log(X_0!)] + \sum_{i=1}^{n} \left(-\theta X_{i-1} + X_i \log(\theta X_{i-1}) - \log(X_i!)\right) \\
&= -\theta \sum_{i=0}^{n-1} X_i + \left(\sum_{i=0}^{n} X_i\right) \log\theta + \sum_{i=1}^{n} X_i \log X_{i-1} - \sum_{i=0}^{n} \log(X_i!)
\end{aligned}$$

The derivative w.r.t. θ

$$\frac{d}{d\theta}\mathcal{L}(\theta) = -\sum_{i=0}^{n-1} X_i + \frac{1}{\theta}\sum_{i=0}^{n} X_i.$$

Therefore, the MLE of θ is

$$\hat{\theta} = \frac{\sum_{i=0}^{n} X_i}{\sum_{i=0}^{n-1} X_i}$$

Part (ii):

$$I(\theta) = -\mathbb{E}\left[\frac{d^2}{d\theta^2}\mathcal{L}(\theta)\right] = -\mathbb{E}\left[-\frac{1}{\theta^2}\sum_{i=0}^{n} X_i\right] = \frac{1}{\theta^2}\mathbb{E}\left[\sum_{i=0}^{n} X_i\right] = \frac{(n+1)\theta}{\theta^2} = \frac{n+1}{\theta}$$

Problem 2

Let $X = (X_1, \ldots, X_n)$, where

$$X_i \overset{iid}{\sim} f_\theta(x_i) = \frac{1}{2}\exp(-|x_i - \theta|),$$

and $\theta > 0$ is the unknown parameter. Find an MLE of θ. (Hint: Consider the cases where n is even and odd separately and find the MLE in terms of the order statistics.)

Solution

Maximizing the objective function

$$\hat{\ell}(\theta) = \sum_{i=1}^{n} \log f_\theta(x_i) = \sum_{i=1}^{n} \log\left[\frac{1}{2}\exp(-|x_i - \theta|)\right] = -n\log 2 - \sum_{i=1}^{n} |x_i - \theta|$$

The MLE of θ is the median of the order statistics.

Problem 3

Given a fixed known integer $r > 1$, let $X = X_{i,j} : i \in [n], j \in [r]$ such that

$$X_{i,j} \sim \mathcal{N}(\mu_i, \sigma^2)$$

and $X_{i,j}$'s are mutually independent. The unknown parameters are $\theta = (\mu_1, \dots, \mu_n, \sigma^2)$, where μ_i can be any real number for all $i \in [n]$ and $\sigma^2 > 0$.

(i) Find the MLE of θ.

(ii) Is the MLE asymptotically unbiased for σ^2 as $n \to \infty$? Can you explain, intuitively, why or why not?

Solution

Part (i):

$$\begin{aligned}
\ell(\mu_i, \sigma^2) &= \sum_{i=1}^{n}\sum_{j=1}^{r} \log\left[\frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x_{i,j}-\mu_i)^2}{2\sigma^2}\right)\right] \\
&= -nr \log\sqrt{2\pi\sigma^2} - \frac{1}{2\sigma^2}\sum_{i=1}^{n}\sum_{j=1}^{r}(x_{i,j}-\mu_i)^2
\end{aligned}$$

The MLE of w.r.t. μ_i:

$$\frac{d}{d\mu_i}\left(-\frac{1}{2\sigma^2}\sum_{i=1}^{n}\sum_{j=1}^{r}(x_{i,j}-\mu_i)^2\right) = \frac{1}{\sigma^2}\sum_{i=1}^{n}\sum_{j=1}^{r}(x_{i,j}-\mu_i) = 0 \quad \Rightarrow \quad \hat{\mu}_i = \frac{1}{r}\sum_{i=1}^{n}\sum_{j=1}^{r} x_{i,j}.$$

Similarly for σ^2:

$$\frac{d}{d\sigma^2}\ell(\mu_i, \sigma^2) = \frac{nr}{2\sigma^2} + \frac{1}{2\sigma^4}\sum_{i=1}^{n}\sum_{j=1}^{r}(x_{i,j}-\mu_i)^2 = 0 \quad \Rightarrow \quad \hat{\sigma}^2 = \frac{1}{nr}\sum_{i=1}^{n}\sum_{j=1}^{r}(x_{i,j}-\hat{\mu}_i)^2.$$

Part (ii): The MLE is not asymptotically unbiased for σ^2 as $n \to \infty$, as the number

Problem 4

Let $X = (X_1, \dots, X_n)$, where

$$X_i \overset{iid}{\sim} \theta f_1(x_i) + (1-\theta) f_2(x_i).$$

Here, f_1 and f_2 are two different known density functions, and $\theta \in [0, 1]$ is the unknown parameter.

(i) Find a sufficient and necessary condition for which the score equation

$$\frac{d}{d\theta} \log f_\theta(x) = 0$$

has a unique solution. Show that if the score equation has a unique solution, it is the MLE.

(ii) Find the MLE when the score equation has no solution.

Solution

Part (i): The score function has a unique solution if and only if f_1 and f_2 are convex functions.

Part (ii): When the score equation has no solution, the MLE is either $\hat{\theta} = 0$ or $\hat{\theta} = 1$.

Problem 5

You have two biased coins (Coin A and Coin B) with unknown probabilities of landing on heads (θ_A and θ_B). You pick one coin at random, flip it 10 times, and record the results (e.g., (5H, 5T)). You repeat this five times for a total of 50 flips. You do not know which coin was used for each set of 10 flips. This "which coin" information is the latent variable ($U_i \in \{A, B\}, i = 1, 2, 3, 4, 5$). The goal here is to use the EM algorithm to estimate the true bias (θ_A and θ_B) for each coin.

(i) Find an explicit expression for the posterior distribution and the expectation

$$q^{(t)}(u) = f_{\theta^{(t-1)}}(u|x) \quad \text{and} \quad \mathbb{E}_{q^{(t)}}[\log f_\theta(x, U)]$$

in the E-step and the update rule

$$\theta^{(t)} = \arg\max_\theta \mathbb{E}_{q^{(t)}}[\log f_\theta(x, U)]$$

in the M-step.

(ii) Given that the observations for the five trials are:

Trial 1 (5H, 5T), Trial 2 (9H, 1T), Trial 3 (8H, 2T), Trial 4 (4H, 6T), Trial 5 (7H, 3T),

write a compute program to find the MLE of $\theta = (\theta_A, \theta_B)$. Plot the posterior probabilities $p_{\theta^{(t-1)}}(u_i = A|x)$, $i = 1, 2, 3, 4, 5$ and the estimates $(\theta_A^{(t)}, \theta_B^{(t)})$ as a function of the number of iterations t.

Solution

Part (i): Using Bayes' theorem

$$q^{(t)}(u) = f_{\theta^{(t-1)}}(u|x) = \frac{f_{\theta^{(t-1)}}(x|u) f_{\theta^{(t-1)}}(u)}{f_{\theta^{(t-1)}}(x)}$$

Part (ii): $\hat{\theta} = (0.58, 0.71)$